Supplementary Material for Physics-Calibrated Simulation of Deformable Tissue for Ultrasound-Guided Robotic Femoral Access

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This document holds the material the main body refers to as Sections S1–S3, Figs. S1–S7 and Tables S1–S9. Fig. S1 is the causal chain the Introduction describes. The videos and the simulation code behind every simulated result, figure and table are in the project repository.[1] Section, figure, equation and table numbers without an S prefix refer to the main body.

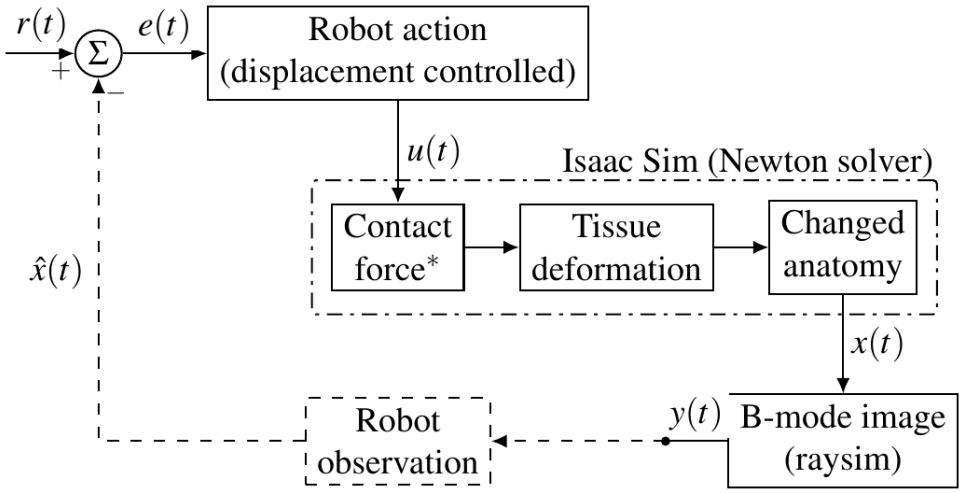


Fig. S1. Causal chain from robot action to robot observation in ultrasound-guided intervention. Every link passes through tissue mechanics, so the physics engine determines what the robot sees. Dashed outline and arrows: not implemented in this work. *The contact force is computed inside the solver but not reported (main body, Section IV-D).

S1. Solver Details

A. Deformation Gradient, Material Parameters, Stress, Per-Vertex Update and Damping

Each tetrahedron has rest vertices $\mathbf{X}_0, \ldots, \mathbf{X}_3$ and current vertices $\mathbf{x}_0, \ldots, \mathbf{x}_3$ (Fig. S2a). Its deformation gradient, which maps the element's rest shape to its current shape, is constant over the element and is computed as [2]

$$\mathbf{F} = \mathbf{D}_s \mathbf{D}_m^{-1}, \tag{1}$$

where $\mathbf{D}_s = [\mathbf{x}_1 - \mathbf{x}_0,\ \mathbf{x}_2 - \mathbf{x}_0,\ \mathbf{x}_3 - \mathbf{x}_0]$ holds the current edge vectors and $\mathbf{D}_m = [\mathbf{X}_1 - \mathbf{X}_0,\ \mathbf{X}_2 - \mathbf{X}_0,\ \mathbf{X}_3 - \mathbf{X}_0]$ holds the rest edge vectors. $\mathbf{D}_m^{-1}$ is computed once from the rest mesh. The main body gives the Stable Neo-Hookean energy density Ψ (its Eq. 1) and the VBD objective $\mathcal{E}(\mathbf{x})$ (its Eq. 2), with $\hat{\mu} = \mu$ and $\hat{\lambda} = \lambda + \mu$. μ is the shear modulus, the material's resistance to change of shape, and λ adds its resistance to change of volume. Both follow from Young's modulus E and Poisson ratio ν (main body, Table II):

$$\mu = \frac{E}{2(1+\nu)}, \qquad \lambda = \frac{E\nu}{(1+\nu)(1-2\nu)}. \tag{2}$$

This mapping makes the energy match linear elasticity at rest [3]. The first Piola–Kirchhoff stress is

$$\mathbf{P} = \frac{\partial \Psi}{\partial \mathbf{F}} = \hat{\mu}\mathbf{F} + \hat{\lambda}(J - \alpha)\,\mathrm{cof}(\mathbf{F}), \tag{3}$$

and vertex i takes the Newton step

$$\mathbf{x}_i \leftarrow \mathbf{x}_i + \mathbf{H}_i^{-1}\mathbf{f}_i, \qquad \mathbf{f}_i = -\frac{\partial \mathcal{E}}{\partial \mathbf{x}_i}, \qquad \mathbf{H}_i = \frac{\partial^2 \mathcal{E}}{\partial \mathbf{x}_i^2}, \tag{4}$$

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[1]https://github.com/MRDLab/femoral-sim-calibration

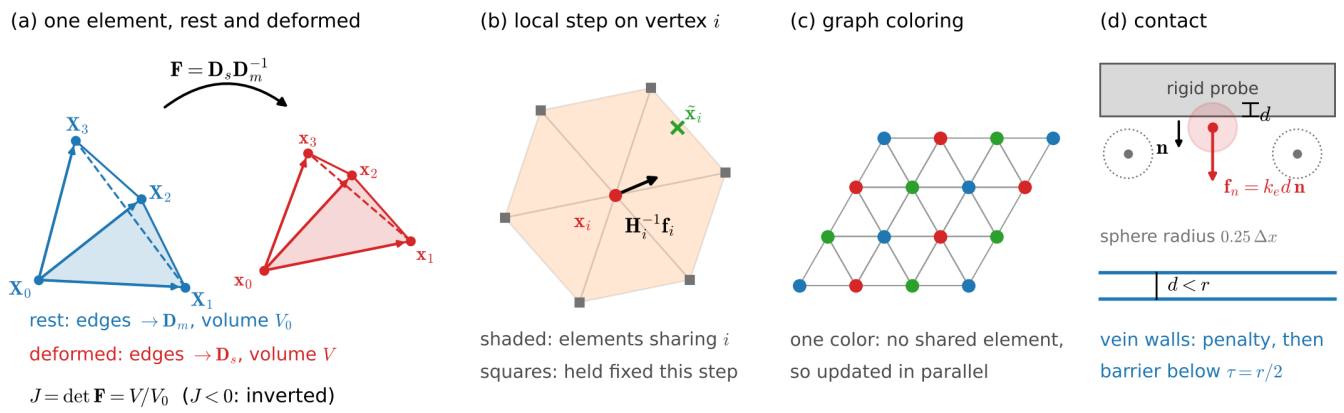


Fig. S2. Vertex Block Descent on a tetrahedral mesh [1]. (a) One element at rest, vertices $\mathbf{X}_0$–$\mathbf{X}_3$, and deformed, vertices $\mathbf{x}_0$–$\mathbf{x}_3$. The edge vectors from vertex 0 form $\mathbf{D}_m$ and $\mathbf{D}_s$, and $\mathbf{F} = \mathbf{D}_s\mathbf{D}_m^{-1}$ maps one to the other. $J = \det \mathbf{F}$ is the deformed volume over the rest volume. (b) A local step: vertex i moves while every other vertex is held fixed, and only the elements that share vertex i (shaded) contribute to its force $\mathbf{f}_i$ and 3×3 Hessian $\mathbf{H}_i$. The inertia term pulls $\mathbf{x}_i$ toward its inertial target $\tilde{\mathbf{x}}_i$. (c) Graph coloring: vertices of one color share no element, so they are updated in parallel on the GPU, one color after another. (d) Contact: a node's sphere that penetrates the probe face by d is pushed out along the face normal $\mathbf{n}$ (main body, Eq. 3). Opposite vessel walls closer than r are held apart by a penalty, then a barrier (Eq. 7).

where only the elements and contacts that touch vertex i enter the sums (Fig. S2b). Vertices that share no element are independent, so a graph coloring groups them and each color is updated in parallel to prevent double counting (Fig. S2c). One VBD iteration is one pass over all colors. A tetrahedron that straddles two materials takes the mean μ and λ of its sub-samples, excluding bone. A negative J marks an inverted element: the Stable Neo-Hookean energy stays defined there [3], so the solver carries such elements without stopping.

The solver version used for every run (Newton 1.2.1) applies a rate-dependent stress

$$\mathbf{P}_{\text{damp}} = \gamma \mathbf{H}\dot{\mathbf{F}}, \qquad \dot{\mathbf{F}} = \frac{\mathbf{F}^{n+1} - \mathbf{F}^n}{\Delta t}, \tag{5}$$

where $\mathbf{H} = \hat{\mu}\mathbf{I} + \hat{\lambda}\,\mathrm{vec}(\mathrm{cof}\,\mathbf{F})\,\mathrm{vec}(\mathrm{cof}\,\mathbf{F})^\top$ is the elastic Hessian with respect to $\mathbf{F}$, without its cofactor-derivative term, and $\gamma = 1.0$ in every element. In this form γ has units of time. The damping stress vanishes when the tissue is at rest, so it shapes the approach to each settled depth. At the end of each time step, VBD updates the stored vertex velocities from the new and previous positions [1]. These velocities carry the tissue's momentum into the next step's inertial target:

$$\mathbf{v}_i^{n+1} = \frac{\mathbf{x}_i^{n+1} - \mathbf{x}_i^n}{\Delta t}. \tag{6}$$

B. Contact

The probe–tissue penalty spring is the main body's Eq. (3). The solver also adds damping while a node approaches the probe, and Coulomb friction. Contact between opposite vessel walls uses a different law. For two wall points a distance $\delta < r$ apart,

$$\frac{\partial \mathcal{E}_{\text{wall}}}{\partial \delta} = \begin{cases} -k_e(r - \delta), & \tau \le \delta < r, \\ -k_e \dfrac{\tau^2}{\delta}, & \delta < \tau, \end{cases} \tag{7}$$

where $r = 0.4\times$ element size is the contact radius and $\tau = r/2$. The first branch is a quadratic penalty while the second is a logarithmic barrier that stops the walls passing through each other. The probe contact force on each node is available from the energy term, and the reaction force can be derived from it but is not reported (main body, Section IV-D).

C. Meshes and Probe Motion

Fig. S3 shows the leg cross-section, the slab's footprint and the probe motion. Skin displacement and lumen area were the primary mechanical outputs: on the slab, lumen area is the polygon area through a ring of 16 lumen-wall nodes. On the leg, the lumen cross-section cut by the plane $s = 20$ mm.

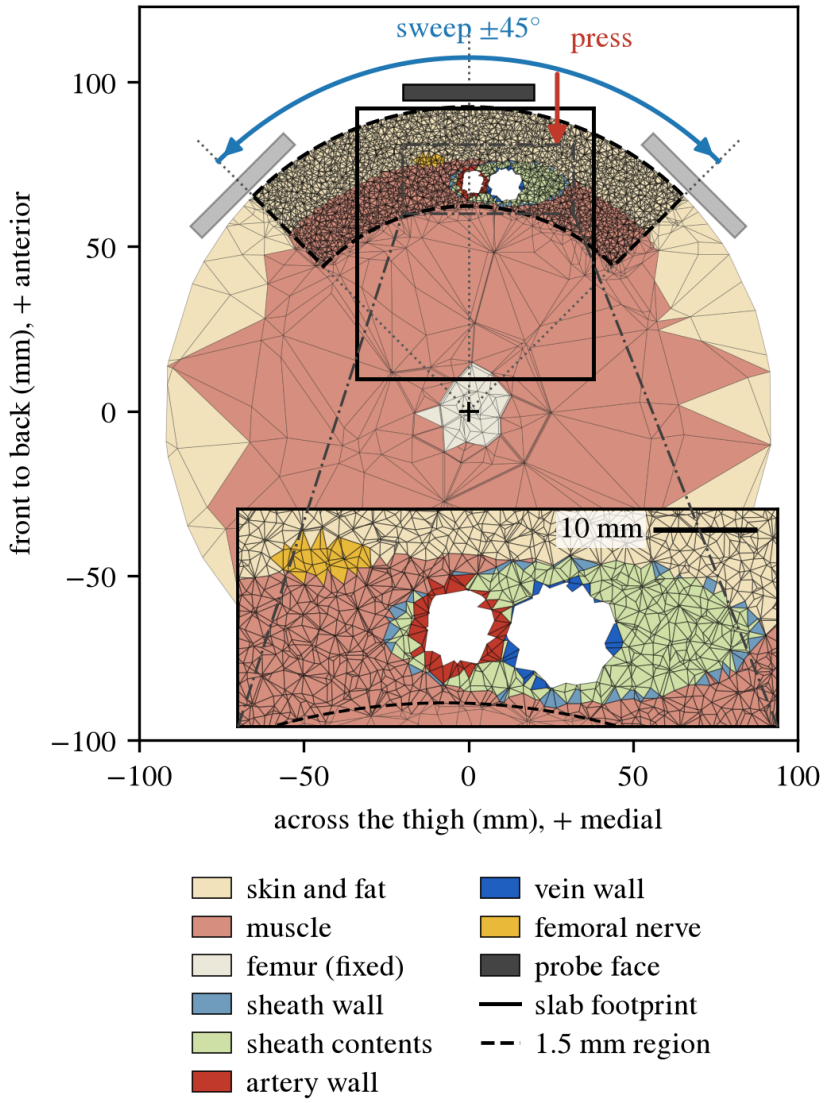


Fig. S3. The leg mesh (129,783 tetrahedra), cut 20 mm below the inguinal ligament, seen from the feet: medial (inner thigh) to the right, anterior (superficial) up. The solid rectangle is the footprint of the slab mesh (166,386 tetrahedra), a rectangular cut of this cross-section on a regular grid. Common femoral artery, vein and nerve sit at the same positions in both meshes, to within 0.55 mm. The dashed outline is the 1.5 mm region: $\pm 45°$ about the front of the thigh, skin to 30.1 mm deep. Mesh edges are drawn at 0.3 pt inside it and 0.2 pt elsewhere. The inset (dash-dot window) enlarges the region around the femoral sheath, showing the artery and vein walls, the sheath wall and its contents, and the femoral nerve. The lumens are voids. The probe (40 mm face, 2 mm off the skin) sweeps $-45°$ to $+45°$ about the limb axis (cross), then presses at $0°$ to 25.4 mm in seven depths. During the sweep only the drawn probe rotates. The contact body stays at $0°$, so the sweep changes only the image.

S2. Slab Studies and Element Inversion

A. *Element Size*

At a 0.0186 ms time step, vein closure at depth still changed with element size (Fig. S4, Table S1): at 25.4 mm, from 33.8% of rest at 1.5 mm to 19.5% at 1.0 mm. The 1.5 mm size was chosen on cost, although a finer mesh still changed the result. At 25.4 mm its vein reading stayed 11.2 points above its closure floor. No inverted element lay in a vessel wall, or within 21.5 mm of one through 15 mm (Section S2-C). Deeper slab readings are used only to compare settings.

Table S1 and Fig. S4 give the element-size study summarized in the main body's Section III-B (0.0186 ms time step, 896 steps per frame, 140 iterations per step). The 1.5 mm size was the cheapest mesh whose contact spring was resolved at 0.0186 ms, where the limit is about 0.026 ms. Element count barely changed the cost per solver pass: 0.677, 0.682, 0.680, 0.688 and 0.745 ms for 2.5, 2.0, 1.5, 1.25 and 1.0 mm (three runs each, alone on the GPU, regenerated by the repository code). The wall clock per press followed the number of frames each depth needed to settle. Without CUDA Graphs each

TABLE S1

ELEMENT-SIZE STUDY ON THE SLAB, COST AND VALIDITY: 0.0186 MS TIME STEP, 140 ITERATIONS PER STEP, NO BLOOD PRESSURE INSIDE THE VESSELS. VEIN: SETTLED LUMEN AREA AT 25.4 MM, % OF REST. FLOOR: THE LEAST AREA EACH MESH CAN REPRESENT (HALF THE RESTING LUMEN PERIMETER TIMES THE GAP AT WHICH OPPOSITE WALLS STOP, 0.8× ELEMENT SIZE, OVER THE RESTING AREA; NOMINAL, UNTESTED). HEADROOM: VEIN MINUS FLOOR, IN POINTS; NEAR ZERO, THE READING REFLECTS THE GRID MORE THAN THE TISSUE. INVERTED: ELEMENTS TURNED INSIDE OUT AT 25.4 MM; WORST RATIO: THE SMALLEST ELEMENT VOLUME RELATIVE TO REST (BELOW 0 = INVERTED). WALL CLOCK: ONE SEVEN-DEPTH PRESS, ALONE ON THE GPU. RED: FEWER THAN THE THREE PLANNED RUNS; N SHOWN.

Element	Tets	Vein	Floor	Headroom	Inverted	Worst ratio	Wall clock
2.5 mm	32,868	57.8±1.3%	35.6%	22.2±1.3	119±14	-2.14±0.57	483.3±14.1 min
2 mm	67,872	33.8±0.3%	29.7%	4.1±0.3	152±8	-1.64±1.24	549.2±9.2 min
1.5 mm	166,386	33.8±0.3%	22.6%	11.2±0.3	304±18	-1.23±0.70	582.2±30.6 min
1.25 mm	284,700	26.0±0.1%	19.3%	6.7±0.1	423±14	-1.44±1.14	486.4±7.6 min
1 mm	544,320	19.5±0.0%	15.8%	3.7±0.0	500±16	-1.68±0.35	590.6±3.0 min

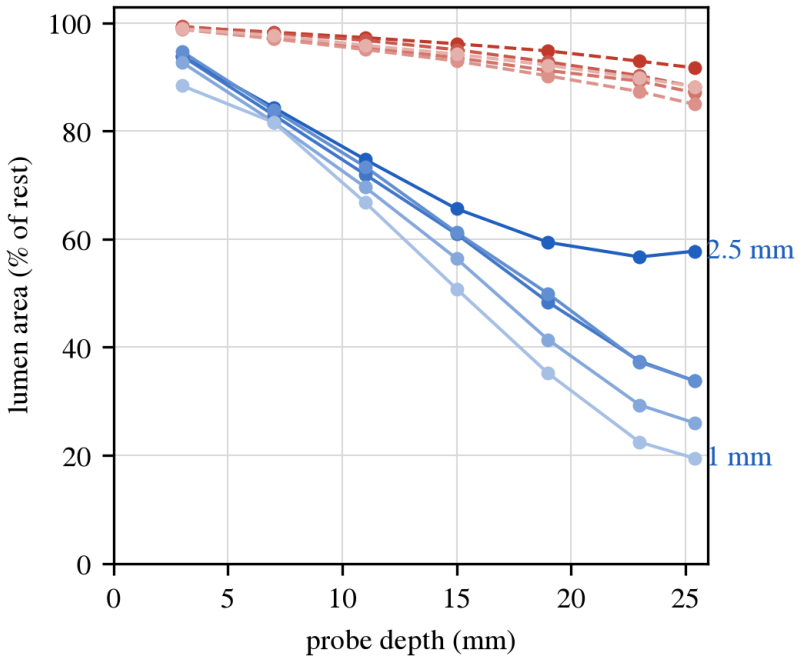


Fig. S4. Slab: effect of element size at a 0.0186 ms time step and 140 iterations per step. Vein (solid) and artery (dashed) lumen area against probe depth for five element sizes, mean of three runs. Lighter is finer: 1 mm lightest, 2.5 mm darkest. Numbers, inverted elements and each mesh's closure floor are in Table S1.

pass is a fixed series of kernel launches. we took that fixed cost to dominate, and it was not measured separately on the slab. Finer elements still cost more in practice, because they need a smaller time step for the contact spring.

B. Time Step and Iterations

Fig. S5 and Table S2 give the time-step study on the 1.5 mm slab. The first settings were far from interactive. At 0.099 ms with 140 iterations, each frame took 16,300 ms alone on the GPU without CUDA Graphs, a seven-depth press of 429 to 431 frames took 116.9 ± 1.2 min, and the 0.0186 ms setting 582.2 ± 30.6 min per press (three runs each). Coarsening to 2.08 ms with 35 iterations gave 335 ± 6 ms per frame and 2.4 ± 0.1 min per press, including per-frame diagnostics a live loop does not need. Profiling a 329 ms frame of this setting on an earlier version of the slab mesh split it into 105 ms of Python dispatch of about 11,000 kernel launches, 91 ms of GPU solver work, 131 ms in two diagnostic calculations (a force estimate and an element-quality check) and under 1 ms of data readback. Solver work per frame is time steps per frame × iterations per step. For the leg reference, it is $168 \times 140 = 23,520$, about 1.6 million kernel launches per frame. Profiled on the slab at a 5.56 ms time step with 35 iterations, the host took about 9.5 μs to issue each launch (9.2 μs on an earlier leg mesh), against 6–8 μs of GPU work in it.

Coarsening the time step 448-fold, to 8.33 ms, moved vein closure by 1.89 ± 0.02 points over three runs (Fig. S5, Table S2). Two settings failed to settle at some depths: their time steps, 0.249 and 0.980 ms, are 0.96 and 3.80 times the contact spring's 0.258 ms period. The 0.980 ms step deviated most, 5.72 ± 0.07 points, while the 0.249 ms step deviated 2.60 ± 0.27, less than the settled 5.56 ms step (4.67 ± 0.39). At 2.08 ms, 35 and 70 iterations stayed within 0.39 ± 0.05 and

TABLE S2

TIME-STEP STUDY ON THE SLAB, STABILITY: 1.5 MM ELEMENTS, 140 ITERATIONS PER STEP, NO CUDA GRAPHS, NO BLOOD PRESSURE INSIDE THE VESSELS. THE PROBE CONTACT ACTS AS A STIFF SPRING ON EACH SURFACE NODE, WITH PERIOD $T = 2\pi\sqrt{\rho\,\Delta x^3/k_e} = 0.258$ MS ($\rho = 1000$ KG/M^3, Δx = 1.5 MM, k_e = 2,000 N/M). SETTLED: EVERY DEPTH MET THE RULE OF 30 FRAMES WITHIN 2%; NO (k OF 7): k DEPTHS RAN TO THE FRAME CAP WITHOUT SETTLING. MOST FRAMES: THE LARGEST NUMBER OF FRAMES ANY SETTLED DEPTH NEEDED. LARGEST GAP: LARGEST DIFFERENCE FROM THE 0.0186 MS REFERENCE OVER THE SEVEN DEPTHS, IN POINTS. WALL CLOCK: ONE SEVEN-DEPTH PRESS, ALONE ON THE GPU. RED: FEWER THAN THE THREE PLANNED RUNS; N SHOWN.

Time step	Per frame	Step / T	Settled	Most frames	Largest gap	Wall clock
0.0186 ms	896	0.07	yes	127±42	ref.	582.2±30.6 min
0.099 ms	168	0.38	yes	177±3	2.26±0.13	116.9±1.2 min
0.249 ms	67	0.96	**no** (4 of 7)	192±37	2.60±0.27	137.3±9.0 min
0.980 ms	17	3.80	**no** (2 of 7)	167±8	5.72±0.07	31.5±0.6 min
2.08 ms	8	8.07	yes	140±2	1.50±0.03	6.4±0.1 min
5.56 ms	3	21.52	yes	247±0	4.67±0.39	4.6±0.1 min
8.33 ms	2	32.29	yes	137±4	1.89±0.02	2.4±0.1 min

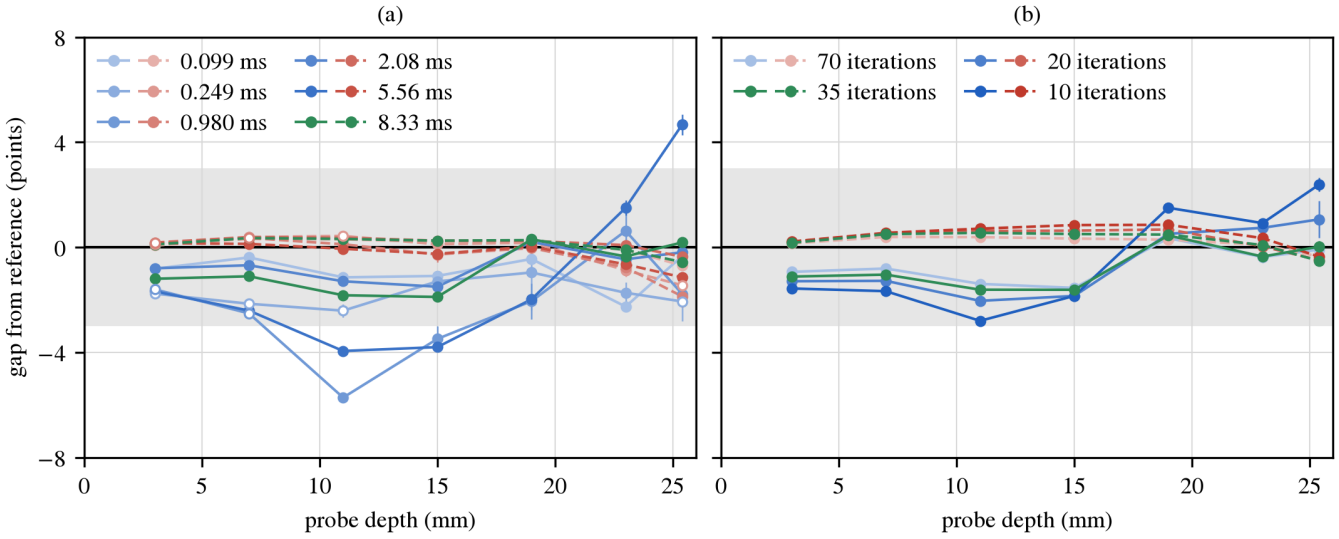


Fig. S5. Slab: difference from the reference run (0.0186 ms time step, 140 iterations per step) in lumen area, in percentage points, against probe depth. Vein solid (blue), artery dashed (red). The black line is the reference and the grey band is ±3 points. (a) Time step varied at 140 iterations per step, lighter to darker from 0.099 to 8.33 ms. (b) Iterations per step varied at a 2.08 ms time step, lighter to darker from 70 to 10. Green: the values carried to the leg model (8.33 ms in (a), 35 iterations in (b)). Each curve is the mean of three runs, and error bars are the SD. Open markers: the depth did not settle in at least one of the three runs (0.249 ms at 3, 7, 11 and 25.4 mm, and 0.980 ms at 3 and 7 mm). The 0.249 and 0.980 ms steps are 0.96 and 3.80 times the contact spring's period. The reference's SD is at most 1.1 points (vein, 23 mm) and otherwise at most 0.3.

0.24 ± 0.07 points of the 140-iteration reference (three runs each). With 10 and 20 iterations, closure drifted 2.58 ± 0.26 and 1.52 ± 0.47 points, so the leg model used 35 iterations per step.

C. Where the Elements Invert

Inverted elements were located in the node positions saved by the six time-step runs on the 1.5 mm slab (regenerated by the repository code). In the four settings that settled at every depth, at most one element was inverted at 11 mm. At 15 mm there were 0 to 6, all within 4.0 mm of the skin under the probe and at least 21.5 mm from either vessel wall. From 19 mm, inversion reached the tissue around the vessels: at 23 and 25.4 mm, 6 to 14 elements between the artery and the vein were inverted in each settled setting. No element of either vessel wall inverted at any depth, and at most 0.97% of the mesh was inverted at 25.4 mm. Whether that region biased every setting equally was not tested. The element-size runs left no saved positions, so other element sizes were not located.

On the leg, inversion was measured in three presses at the chosen setting (regenerated by the repository code). Elements first inverted at 7 mm, all within 1 mm of the skin. Through 15 mm, every inverted element lay at least 15.2 mm from either vessel wall. At 25.4 mm, 811 to 859 elements were inverted, under 0.67% of the mesh. Over 99% of them were skin and fat, and four lay between the artery and the vein. No element of either vessel wall inverted at any depth, and every run stayed finite.

The force-balance check sums the internal vertical force on interior nodes away from the probe. Since nothing but the probe loads them, it should read zero. On the 1.5 mm slab at a 0.0186 ms time step it stayed within 0.08 N through 15 mm

over three runs, with at most five elements inverted, then grew to 0.16, 0.78 and 1.39 N at 19, 23 and 25.4 mm while inverted elements grew to 16, 85 and 304. At 25.4 mm the error was 28% of the 4.9 N force reading. The error and the inversion grew together, although a causal link was not shown. At 15 mm, 32–100% of the inverted elements lay within 3 mm of a probe edge, where its load enters the tissue.

S3. Leg Model: Every Setting, the Benchtop Comparison and the Sensitivity Cases

A. Cost Model and Every Solver Setting

TABLE S3

WHERE THE WALL CLOCK GOES ON THE LEG MODEL, ONE RTX 5090, SOLVER ONLY. TOP: THE COST MODEL FITTED TO EVERY REPEATED RUN, MINIMIZING RELATIVE ERROR; ± IS ONE STANDARD DEVIATION BY BOOTSTRAP OVER RUNS. THE PER-TIME-STEP COST IS WHY TWO SETTINGS WITH EQUAL SOLVER WORK DIFFER IN SPEED. BOTTOM: MEDIAN WALL CLOCK PER FRAME WITH CUDA GRAPHS OFF AND ON (THE GPU WORK RECORDED ONCE AND REPLAYED AS ONE LAUNCH), AT EQUAL SETTINGS; GAIN = OFF / ON. RED: FEWER THAN THE FIVE PLANNED RUNS, OR AWAITING RUNS; N SHOWN.

Cost per simulated frame (60 runs; residuals: mean 0.99%, largest 4.57%)			
per solver pass	0.400 ± 0.001 ms		
per time step	0.20 ± 0.03 ms		
per frame, fixed	2.04 ± 0.16 ms		
Time step, iterations	**Graph off (ms)**	**Graph on (ms)**	**Gain**
8.33 ms, 35	90.4±0.5	30.4±0.1	2.97±0.01×
2.08 ms, 35	358.6±2.3	117.5±1.7	3.05±0.04×
0.099 ms, 140	29701.8±135.6	9301.7±210.9	3.19±0.07×

TABLE S4

LEG MODEL: SPEED AND ACCURACY OF EVERY SOLVER SETTING, CUDA GRAPHS ON, SOLVER ONLY. EVERY FILLED POINT OF FIG. 3 IS A ROW HERE, LABELLED OR NOT; THE HOLLOW POINTS ARE IN TABLE S3. SOLVER WORK (THE FIGURE'S X AXIS): TIME STEPS PER FRAME × ITERATIONS PER STEP. SPEED-UP PER FRAME: THE REFERENCE SETTING'S MS PER FRAME OVER THIS SETTING'S, AS THE FIGURE PLOTS IT. SPEED-UP PER PRESS: WALL CLOCK TO SETTLE ALL SEVEN DEPTHS, AGAINST THE REFERENCE SETTING. FASTER SETTINGS NEED MORE FRAMES TO SETTLE, SO THE PER-PRESS GAIN IS SMALLER. GAP: THE LARGEST DIFFERENCE FROM THE REFERENCE SETTING'S MEAN OVER THE SEVEN DEPTHS, IN PERCENTAGE POINTS OF RESTING AREA. BOLD: ABOVE THE 3-POINT TOLERANCE SET BEFORE THE RUNS. MEAN ± SD OVER FIVE RUNS; RED: FEWER, N SHOWN.

Time step, iterations	Work	ms per frame	Speed-up, frame	Speed-up, press	Vein gap	Artery gap
0.099 ms, 140 (reference)	23,520	9302±211	1	1	0.23±0.16	0.13±0.06
0.249 ms, 140	9,380	3798±14	2.4±0.0	2.5±0.1	0.33±0.09	0.34±0.08
0.980 ms, 140	2,380	953±6	9.8±0.1	9.6±0.3	0.19±0.07	0.25±0.06
2.08 ms, 140	1,120	453±3	20.5±0.1	19.1±1.0	0.24±0.13	0.35±0.19
5.56 ms, 140	420	172±2	54.1±0.6	46.5±2.9	0.89±0.42	0.85±0.00
8.33 ms, 140	280	115±1	81.0±0.7	65.6±5.4	0.58±0.04	0.96±0.05
2.08 ms, 70	560	225±2	41.3±0.4	36.4±0.6	0.41±0.08	0.31±0.04
2.08 ms, 35	280	118±3	79.1±1.8	64.9±6.6	1.20±0.73	0.78±0.00
2.08 ms, 20	160	68.3±0.7	136±1	93.0±1.6	0.87±0.17	0.89±0.08
2.08 ms, 10	80	35.4±0.3	262±2	169.5±7.9	1.10±0.49	1.10±0.23
5.56 ms, 35	105	44.6±0.3	208±1	151.7±4.0	0.85±0.10	1.05±0.12
8.33 ms, 35 (chosen)	70	30.4±0.2	306±2	203.0±2.6	0.76±0.09	1.03±0.00

Table S3 breaks the leg model's wall clock into its parts and gives the CUDA Graph gain at equal settings. Table S4 gives the speed and accuracy of every setting. Fig. S6 gives each setting's departure from the fitted line. Every leg setting settled at every depth in all five runs, so whether the settling check, developed on the slab, can reject a leg setting remains untested. In the full loop (one run of the chosen setting with ray-traced ultrasound and the rendered 3-D scene), a frame took 56.4 ms: the solver 36.6 ms and the 3-D view 15.9 ms, with the ultrasound image computed in parallel. Tables S7 and S8 give every setting's lumen areas at every depth, the chosen setting among them. The artery read slightly above 100% at 3 and 7 mm. Whether this is lateral widening or error of the plane-cut area measurement is undecided.

B. Benchtop Comparison

Table S5 gives the benchtop and simulated lumen areas at every depth, and Fig. S7 every measured image. The supplier's batch spread of the gels, ±1 SD, moved the simulated lumens by about ±1 point at 25.4 mm. The vein discrepancy from 15 mm on most likely comes from the vein tube's wall stiffness relative to the gel: its 35A wall may be stiffer than Gent's relation gives, or the gel softer than its supplier's value, and measuring either material directly would settle it. Two modelling choices changed the result during development: a pinned femur over-constrained the free-floating benchtop phantom and cost about 10 points of vein area, and the self-contact radius had to be sized to the mesh, or refined tube walls did not move. The lumen measurement script (`measure_lumen.py`) and the operator's picks are in the repository.

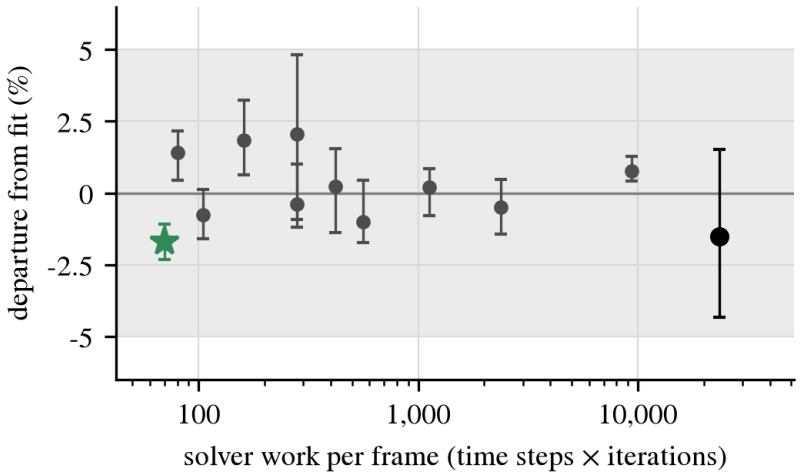


Fig. S6. Departure of each CUDA-Graphs-on setting from the two-term fit drawn in the main body's speed figure, 100 × (measured − fit)/fit. Point: mean of five runs, bar: fastest to slowest run, black: reference, star: chosen setting, grey band: ±5%. Mean absolute departure 1.26%, largest 4.83%. The two-term fit is 0.401 ms per solver pass plus 2.82 ms per frame (bootstrap SD 0.001 and 0.19 ms). The three-term model of Table S3 adds a per-time-step cost, so its fixed cost (2.04 ms) and residuals (0.99% mean, 4.57% largest) differ.

TABLE S5

BENCHTOP PHANTOM AGAINST THE SIMULATED PHANTOM: LUMEN AREA, % OF REST. BENCHTOP: MEAN ± SD OF THREE PRESSES (THE THIRD 25.4 MM PRESS FROM A SEPARATE SESSION). SIMULATED: THE SIMULATED PHANTOM, REFERENCE SETTING. ERROR: SIMULATED MINUS BENCHTOP, IN PERCENTAGE POINTS; BOLD: OUTSIDE THE 5-POINT TOLERANCE SET BEFORE THE TRIALS. WITHIN 5 POINTS: ARTERY 7 OF 7 DEPTHS, VEIN 3 OF 7.

	Vein			Artery		
Depth	bench.	sim.	error	bench.	sim.	error
3 mm	95.8±1.9	95.9	+0.1	93.3±2.9	94.2	+0.8
7 mm	90.1±0.8	89.0	-1.1	87.2±0.9	87.8	+0.6
11 mm	84.0±0.3	81.0	-3.0	77.9±1.8	80.8	+2.9
15 mm	78.5±0.4	72.3	**-6.1**	71.1±3.1	73.5	+2.4
19 mm	70.6±2.6	63.0	**-7.6**	67.2±3.4	65.9	-1.3
23 mm	65.0±4.1	53.1	**-11.9**	58.8±2.8	57.2	-1.6
25.4 mm	52.3±0.9	47.1	**-5.3**	47.0±1.6	51.6	+4.6

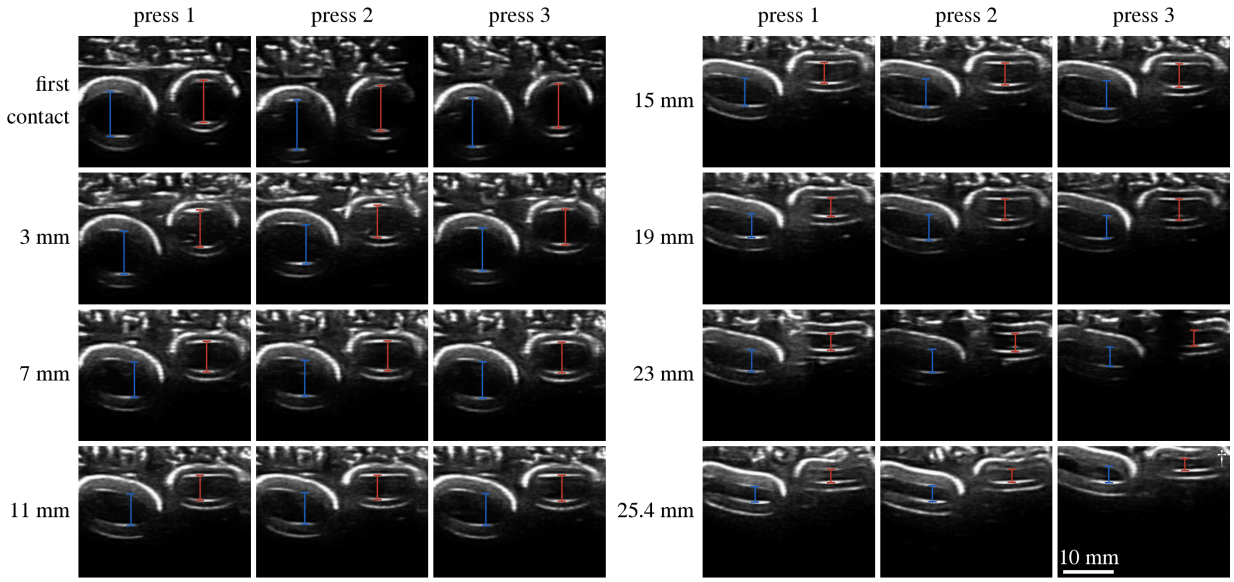


Fig. S7. Every benchtop phantom image behind the main body's benchtop comparison, one crop (34 × 26 mm) at one scale (bar: 10 mm). Rows: first contact and the seven press depths, columns: the three presses at each depth. Caliper lines are the measured inner heights: vein (left tube) blue, artery (right tube) red. †The third 25.4 mm image was taken in a separate session. Its tubes sit about 4 mm higher than in presses 1 and 2.

C. Vein-Wall and Nerve Stiffness

TABLE S6

LEG MODEL: SENSITIVITY OF THE VESSEL RESPONSE TO THE VEIN-WALL AND NERVE STIFFNESS. ONE MATERIAL IS CHANGED PER CASE; EVERYTHING ELSE IS IDENTICAL (8.33 MS TIME STEP, 35 ITERATIONS PER STEP, CUDA GRAPHS ON, SOLVER ONLY). THE CONTROL IS THE PAPER'S MODEL. BENDING MATCH: THE MODULUS AT WHICH A ONE-ELEMENT (1.5 MM) WALL BENDS LIKE A REAL WALL OF THAT THICKNESS AT 5,700 KPA, $E_{\mathrm{eff}} = 5{,}700\ \mathrm{kPa} \times (t/1.5\ \mathrm{mm})^3$ [4], [5]. MEAN ± SD OVER FIVE RUNS; RED: FEWER, N SHOWN. $^{\dagger k}$NOT SETTLED IN k OF THE REPEATS: THE AREA WAS STILL CHANGING AT THE END OF THE SETTLING ALLOWANCE, AND THE VALUE IS THE LAST FRAME. A FULLY FLATTENED VEIN WOULD READ ABOUT 14–20% ON THIS MESH (SECTION III-D).

Vein wall (or nerve) modulus	**3 mm**	**7 mm**	**11 mm**	**15 mm**	**19 mm**	**23 mm**	**25.4 mm**
Vein lumen area, % of rest							
20 kPa, as in Table II (control)	96.5±0.0	87.8±0.0	75.4±0.0	62.1±0.0	48.0±0.1	33.7±0.2	27.6±0.1
20 kPa, uniform	96.7±0.0	88.5±0.0	76.5±0.0	63.6±0.0	49.6±0.1	34.7±0.2	27.8±0.2
66.4 kPa: bending match, 0.34 mm wall	97.8±0.0	91.8±0.0	81.8±0.0	71.1±0.0	58.2±0.1	45.7±0.2	35.3±0.5
211.1 kPa: bending match, 0.50 mm wall	98.8±0.0	95.5±0.0	88.1±0.0	80.4±0.1	71.5±0.1	60.3±0.1	51.4±0.2
600 kPa (the artery wall's value)	99.5±0.0	98.3±0.0	93.3±0.0	88.3±0.0	82.8±0.0	76.0±0.1	70.3±0.0
5,700 kPa (tensile value, as a solid wall)	99.9±0.0	100.0±0.0	100.5±0.0	98.7±0.0	96.1±0.0	94.1±0.0	93.0±0.1
nerve 20 kPa (half)	96.5±0.0	87.8±0.0	75.5±0.0	62.3±0.0	47.9±0.0	33.7±0.2	27.6±0.1
nerve 80 kPa (double)	96.5±0.0	87.9±0.0	75.3±0.0	62.2±0.0	47.9±0.1	34.0±0.2	27.8±0.2
Artery lumen area, % of rest							
20 kPa, as in Table II (control)	100.6±0.0	100.9±0.0	99.7±0.0	92.9±0.0	94.1±0.0	89.9±0.1	83.6±0.1
20 kPa, uniform	100.7±0.0	100.9±0.0	99.7±0.0	92.9±0.0	94.1±0.0	90.0±0.0	83.8±0.1
66.4 kPa: bending match, 0.34 mm wall	100.7±0.0	101.0±0.0	99.9±0.0	93.0±0.0	94.3±0.0	90.5±0.0	84.8±0.2
211.1 kPa: bending match, 0.50 mm wall	100.6±0.0	101.1±0.0	100.2±0.0	93.3±0.0	94.8±0.0	91.2±0.0	86.2±0.1
600 kPa (the artery wall's value)	100.5±0.0	101.4±0.0	100.7±0.0	94.0±0.0	95.7±0.0	92.2±0.0	87.5±0.1
5,700 kPa (tensile value, as a solid wall)	99.8±0.0	100.4±0.0	101.1±0.0	100.4±0.0	94.1±0.1	96.8±0.0	91.6±0.1
nerve 20 kPa (half)	100.7±0.0	100.9±0.0	99.7±0.0	93.0±0.0	94.2±0.1	89.8±0.1	83.5±0.2
nerve 80 kPa (double)	100.6±0.0	100.9±0.0	99.8±0.0	92.9±0.0	94.1±0.0	90.3±0.1	83.7±0.1

Table S6 gives the sensitivity cases discussed in the main body's Limitations. At 5,700 kPa [4], a one-element (1.5 mm) wall is 27 to 86 times too stiff in bending for a 0.33–0.35 mm wall [5]. A solid 5,700 kPa wall kept 93.0% of the vein's area at 25.4 mm, although that case carries the bending overstatement. A 600 kPa wall kept 70.3% and settled at every depth. Linear tetrahedra are also too stiff in bending, so every stiffened case is biased against flattening. The press sequence ends at 25.4 mm (one inch) by design, and deeper pressing was not simulated.

Published stiffness values, quoted as published, for comparison with the main body: in vivo thigh compression fitted by inverse finite elements gave median shear moduli of 0.537 kPa for fat and 1.675 kPa for muscle [6], at 7.8 mm/s, already described as "quite high for quasi-static assumption". Our probe advances 24 mm/s, about three times that rate. Magnetic resonance elastography of the thigh gives 3.04 kPa for subcutaneous fat [7] and 3.7 to 7.5 kPa for muscle, the upper value the sartorius alone, the two vasti 3.7 to 3.9 kPa, at 90 to 120 Hz [7], [8]. In Young's modulus, subcutaneous fat in tension had an initial value of 1.6 kPa ex vivo [9], breast fat 3.25 kPa [10], and forearm hypodermis about 15 kPa at 0.2–1 kHz [11]. In vivo vastus lateralis gave shear moduli of 5.77 kPa along its fibers and 1.93 kPa across them [12]. Femoropopliteal arteries from 472 donors held a circumferential stiffness near 750 kPa [13], tested ex vivo below the common femoral segment. Autopsy adventitia is anisotropic and strongly nonlinear [14]. Femoral vein from 41 body donors gave 5,700 kPa along the vessel [4]. Saphenous vein is highly compliant over the venous pressure range, then stiffens sharply [15], [16]. Shear-wave elastography measures the femoral nerve along its fibers [17]. Over the femoral nerve of a supine subject shear waves travelled at 2.87 m/s [18], rising with limb posture and probe alignment [19]. Fascia lata from cadavers aged 75–92 gave 71,600–275,900 kPa along the thigh and 3,200–41,900 kPa across it [20]. No study of the femoral sheath itself, or of the tissue inside it, was found.

D. Acoustic Properties

The seven fixed acoustic materials of the Isaac for Healthcare simulator [21] (impedance, attenuation, speed of sound, scattering) were shipped with its liver workflow, and none was measured for the thigh. Tissues without their own entry borrow one: the artery wall, vein wall and femoral sheath use muscle, while the femoral nerve and the tissue inside the sheath use fat. Every elastic modulus is ours, since the simulator supplies none.

TABLE S7

LEG MODEL, VEIN LUMEN AREA AS A PERCENTAGE OF REST AT EACH COMMANDED PROBE DEPTH, FOR EVERY SOLVER SETTING TESTED: TIME STEP AND ITERATIONS PER STEP, CUDA GRAPHS ON, SOLVER ONLY, NO BLOOD PRESSURE INSIDE THE VESSELS. MEAN ± SD OVER REPEATED RUNS; RED: FEWER THAN THE FIVE PLANNED RUNS, N SHOWN.

Time step, iterations	**3 mm**	**7 mm**	**11 mm**	**15 mm**	**19 mm**	**23 mm**	**25.4 mm**
0.099 ms, 140 (reference)	97.1±0.0	88.1±0.0	75.2±0.0	61.7±0.0	47.4±0.1	34.3±0.3	28.4±0.2
0.249 ms, 140	97.1±0.0	88.1±0.0	75.2±0.0	61.4±0.0	47.2±0.2	34.1±0.1	28.2±0.1
0.980 ms, 140	97.1±0.0	88.1±0.0	75.1±0.0	61.7±0.0	47.5±0.1	34.3±0.2	28.3±0.1
2.08 ms, 140	97.1±0.0	88.1±0.0	75.3±0.0	61.6±0.0	47.3±0.2	34.3±0.1	28.2±0.1
5.56 ms, 140	96.9±0.0	88.1±0.0	75.5±0.0	62.4±0.0	47.5±0.1	34.5±0.8	28.4±0.2
8.33 ms, 140	96.7±0.0	87.8±0.0	75.5±0.0	62.2±0.0	47.2±0.1	34.2±0.1	28.1±0.4
2.08 ms, 70	97.0±0.0	88.0±0.0	75.1±0.0	61.5±0.0	47.1±0.2	34.5±0.1	28.4±0.1
2.08 ms, 35	97.0±0.0	88.1±0.0	75.7±0.0	62.3±0.1	47.5±0.1	34.6±1.3	28.2±0.8
2.08 ms, 20	97.0±0.0	87.8±0.0	75.0±0.0	61.6±0.1	47.4±0.2	34.1±0.1	27.5±0.2
2.08 ms, 10	96.9±0.0	87.9±0.0	75.3±0.0	61.7±0.0	47.8±0.1	33.6±0.6	27.3±0.5
5.56 ms, 35	96.8±0.0	88.0±0.0	75.6±0.0	62.3±0.0	47.8±0.1	34.3±0.6	27.6±0.2
8.33 ms, 35 (chosen)	96.5±0.0	87.8±0.0	75.4±0.0	62.1±0.0	48.0±0.1	33.8±0.2	27.7±0.1

TABLE S8

LEG MODEL, ARTERY LUMEN AREA AS A PERCENTAGE OF REST AT EACH COMMANDED PROBE DEPTH, FOR EVERY SOLVER SETTING TESTED: TIME STEP AND ITERATIONS PER STEP, CUDA GRAPHS ON, SOLVER ONLY, NO BLOOD PRESSURE INSIDE THE VESSELS. MEAN ± SD OVER REPEATED RUNS; RED: FEWER THAN THE FIVE PLANNED RUNS, N SHOWN.

Time step, iterations	**3 mm**	**7 mm**	**11 mm**	**15 mm**	**19 mm**	**23 mm**	**25.4 mm**
0.099 ms, 140 (reference)	101.6±0.0	101.1±0.0	98.7±0.0	93.3±0.0	93.9±0.0	90.4±0.0	84.6±0.2
0.249 ms, 140	101.5±0.0	101.1±0.0	98.7±0.0	93.0±0.0	93.9±0.0	90.4±0.1	84.9±0.2
0.980 ms, 140	101.4±0.0	101.0±0.0	98.8±0.0	93.4±0.0	93.9±0.0	90.5±0.1	84.4±0.1
2.08 ms, 140	101.3±0.0	101.0±0.0	98.9±0.0	93.2±0.3	93.9±0.0	90.4±0.0	84.5±0.4
5.56 ms, 140	101.1±0.0	100.9±0.0	99.6±0.0	92.7±0.0	94.0±0.0	89.9±0.1	84.6±0.2
8.33 ms, 140	101.0±0.0	100.8±0.0	99.6±0.0	93.1±0.0	94.0±0.1	89.4±0.1	84.4±0.2
2.08 ms, 70	101.5±0.0	101.1±0.0	98.8±0.0	93.0±0.0	93.9±0.0	90.3±0.1	84.5±0.2
2.08 ms, 35	101.6±0.0	101.1±0.0	99.5±0.0	92.9±0.0	94.1±0.1	89.8±0.1	84.8±0.3
2.08 ms, 20	101.8±0.0	101.2±0.0	98.2±0.0	93.5±0.0	93.9±0.0	90.3±0.1	83.7±0.1
2.08 ms, 10	101.7±0.0	101.2±0.0	98.1±0.0	93.3±0.0	94.0±0.0	90.0±0.2	83.5±0.2
5.56 ms, 35	101.2±0.0	101.0±0.0	99.7±0.0	92.9±0.0	94.1±0.0	89.9±0.1	83.7±0.2
8.33 ms, 35 (chosen)	100.6±0.0	100.9±0.0	99.7±0.0	92.9±0.0	94.1±0.0	90.0±0.0	83.7±0.1

VIDEOS

Every clip shows one live run on the V5 leg mesh: the sweep, then the seven-depth press. The 3-D view is on the left and the ray-traced B-mode image on the right. Each is a single run, shown for illustration. The quantitative results come from the five-run repeats. The B-mode uses the default acoustic materials and has not been compared with clinical images (main body, Section IV-D). The dark columns below the vessels are not acoustic shadowing. Each ray stops after 15 surface crossings, the ray tracer's default, and the vessel walls, lumens and sheath exceed that. In a still frame with the limit raised to 30, the columns vanished and the tissue below the vein was 1.4 dB brighter than its surroundings, as blood's low attenuation predicts. No lumen area depends on the image: every one is measured on the tissue mesh. No clip plays in real time: each frame advances 16.7 ms of simulated time but takes longer to compute.

- `clips/submission/`: the submission video. One 3-D view of the chosen setting beside the ultrasound images of the chosen and intermediate settings, on one clock of computer time. Neither setting runs in real time.
- `clips/final_run/`: the chosen setting (8.33 ms time step, 35 iterations), run 2026-10-21. Each frame is shown at the rate it was computed: 56.4 ms median interval, about 18 video frames per second. Two cut-away runs show the tissue inside the leg. They run at 13 to 14 frames per second and are not quoted for speed.
- `clips/intermediate/` and `clips/longer/`: two slower settings of Table S4, run 2026-11-05. Intermediate: 5.56 ms time step, 140 iterations. Longer: 2.08 ms time step, 140 iterations. The chosen setting was re-run the same day and took 56.3 ms per frame.
- `comparison/`: the chosen setting (top) and the slower setting (bottom) on one shared clock of computer time, played at normal speed. Each panel shows only the newest 3-D view and ultrasound image the computer had finished at that moment. Nothing is resampled or interpolated.
- `alone/`: the slower setting by itself, each frame at the rate it was computed (5 and 2 video frames per second).

TABLE S9

SPEED OF THE RUNS IN THE COMPARISON CLIPS (2026-11-05, ONE RUN EACH). SPEED IS SIMULATED MOTION OVER THE TIME TO ITS LAST IMAGE. EACH DEPTH IS HELD UNTIL IT SETTLES, SO THE SIMULATED MOTION DIFFERS BETWEEN SETTINGS.

Setting	Median frame interval (ms)	Simulated motion (s)	Last image (s)	Speed (× real time)
Chosen (8.33 ms, 35 iterations)	56.3	11.20	38.1	0.29
Intermediate (5.56 ms, 140 iterations)	192.4	9.18	106.4	0.086
Longer (2.08 ms, 140 iterations)	462.0	8.98	252.4	0.036

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